- Due Feb 6 at 11:59pm
- Points 10
- Questions 13
- Time Limit None

Instructions

Week 05

Pre-lab Quiz

Instructions

Review the following videos, and all the study materials in the module then answer the questions.

- **Protocols: UART, I2C, SPI (https://www.youtube.com/watch?v=lyGwvGzrqp8&ab_channel=Electronoobs)**
- **Understanding I2C (https://www.youtube.com/watch?v=CAvawEcxoPU)**
- **Understanding SPI (https://www.youtube.com/watch?v=0nVNwozXslc)**

Attempt History

	Attempt	Time	Score
LATEST	Attempt 1	19 minutes	7.5 out of 10

Score for this quiz: 7.5 out of 10

Submitted Feb 4 at 10:37am

This attempt took 19 minutes.

Correct answer

Question 1

0.5 / 0.5 pts

Which of the following statements is true?

- I2C communication typically exceeds 1 MHz in standard applications
- SPI is limited to half-duplex communication only
- UART and SPI interfaces are always full-duplex
- UARTs can operate at any speed as long as the clock is configured

- SPI can be configured to operate at speeds of 30 MHz or higher

SPI (Serial Peripheral Interface) is a high-speed, synchronous communication protocol that can be configured to operate at 30 MHz or even higher, depending on the hardware. This makes it suitable for applications requiring fast data exchange. In contrast, I2C typically operates at lower speeds, with standard modes at 100 kHz or 400 kHz and high-speed modes up to 3.4 MHz, which are less commonly used. UARTs are limited by factors such as clock accuracy, noise, and hardware capabilities, and not all UART or SPI implementations support full-duplex communication.

Wrong answer

Question 2

0 / 0.5 pts

Which of the following uses a parity bit to check the integrity of transmitted data?

- None of the above
- I2C
- SPI
- All of the above
- UART

UART (Universal Asynchronous Receiver/Transmitter) is the only protocol among the options that can use a parity bit for basic error detection. The parity bit is optional and can be configured to detect single-bit errors in transmitted data. I2C and SPI do not include parity bits in their standard protocol specifications; instead, they rely on other mechanisms or higher-level protocols for error detection and correction. This makes UART unique in its built-in support for parity-based integrity checking.

Correct answer

Question 3

0.5 / 0.5 pts

Which of the following can have the fastest transfer speed?

- UART with flow control
- UART
- I2C with clock stretching
- I2C
- SPI

The correct answer is **SPI**. Among the listed interfaces, SPI (Serial Peripheral Interface) typically supports the highest data transfer rates, often reaching tens of megabits per second depending on the microcontroller and peripheral. I2C is slower, usually maxing out at 1 Mbps (Fast-mode) or 3.4 Mbps (High-speed mode), and UART is generally limited to lower speeds (commonly up to 1 Mbps or slightly

higher with special configurations). SPI's full-duplex nature and lack of addressing overhead also contribute to its speed advantage.

Correct answer

Question 4

0.5 / 0.5 pts

Looking at the datasheet for the STM32 (in the module), what is the **minimum output high voltage** for a normal CMOS I/O pin with a loading of 8mA and input voltage (VDDIOX) of 3.3V?

- 1.8V
- 0.4V
- 2.4V
- 2.9V
- 3.3V

The correct answer is **2.9V**. For CMOS outputs on STM32 microcontrollers operating at 3.3V, the minimum output high voltage (VOH) under an 8mA load is typically around 2.9V. This ensures compatibility with CMOS input thresholds, which require the high level to be close to the supply voltage. The 2.4V value is more typical of TTL logic levels and is not sufficient for CMOS signaling at 3.3V. The 0.4V value refers to the maximum output low voltage (VOL), not VOH.

Correct answer

Question 5

0.5 / 0.5 pts

Which of the following communication interfaces requires the master to send an address before transmitting data, and only the slave that recognizes the address will respond?

- SPI
- UART with parity
- UART
- SPI with multiple chip selects
- I2C

In I2C communication, the master device initiates communication by sending a 7- or 10-bit address that identifies the target slave device. Only the slave with a matching address responds, allowing multiple devices to share the same bus. In contrast, **SPI** uses dedicated chip select lines to address slaves, and **UART** is typically point-to-point and does not use addressing at the protocol level.

Correct answer

Question 6

0.5 / 0.5 pts

Match each communication protocol to a characteristic feature of that protocol:

I2C

Uses a passive signaling sys

SPI

Requires mutually-exclusive

UART

Typically connects only two d

- **I2C** uses open-drain (passive) signaling on shared SDA (data) and SCL (clock) lines, allowing multiple devices to communicate over the same two wires. Devices are addressed individually, and pull-up resistors are required.
- **SPI** requires a separate chip select (CS) line for each slave device, ensuring that only one slave communicates with the master at a time. This makes the peripherals mutually exclusive.
- **UART** is a point-to-point protocol, typically connecting just two devices (e.g., a microcontroller and a terminal) using TX and RX lines. It does not support addressing or shared bus configurations.

Correct answer

Question 7

1 / 1 pts

Why does I2C not require extra signals to select the appropriate slave for a transaction?

- (selected) Because each slave device has a unique address, and the master selects the slave by sending its address over the bus
- Because the master and slave are pre-configured to communicate at specific times
- Because the slave devices take turns responding in a round-robin fashion
- Because the master uses a dedicated chip select line for each slave
- Because I2C uses a broadcast mode to communicate with all slaves at once

I2C uses a unique 7-bit or 10-bit address for each slave device on the bus. When the master initiates communication, it sends the address of the intended slave over the shared data line (SDA). Only the slave with the matching address responds, eliminating the need for separate chip select (CS) lines like those used in SPI. This addressing scheme allows I2C to support multiple devices with just two wires.

Wrong answer

Question 8

0 / 1 pts

What does the TX FIFO empty event indicate?

- The receive FIFO is full and must be read.
- The UART peripheral has been disabled.
- The baud rate has changed and needs to be reconfigured.
- The last byte has been fully transmitted on the wire.
- The transmit FIFO has space available for new data to be written.

The TX FIFO Empty (TXE) event indicates that the transmit FIFO buffer has room for more data—it is either completely empty or has dropped below a threshold. This event is commonly used in interrupt-driven or DMA-based UART transmission to signal that the software or DMA controller can write additional bytes to the FIFO. It does not mean that all data has been sent on the wire (that's the Transmission Complete (TC) flag); it simply means the UART is ready to accept more data for transmission.

Correct answer

Question 9

1 / 1 pts

What four SPI settings are predefined and must be known in advance for both the master and the slave?

- Address, data width, clock skew, and sync pulse
- Clock speed, parity, stop bits, and data length
- Start bit, stop bit, parity, and baud rate
- Clock polarity, clock phase, packet length, and bit order
- Baud rate, voltage level, data rate, and parity

For SPI communication to function correctly, both the master and the slave must be configured with the same four key settings: clock polarity (CPOL), clock phase (CPHA), packet length (number of bits per transfer), and bit order (MSB-first or LSB-first). These settings determine how data is synchronized and interpreted on the bus. Mismatched configurations between devices can lead to corrupted or misaligned data.

Wrong answer

Question 10

0 / 1 pts

How many master devices can be present on an I2C bus at one time?

- One master per slave device
- Unlimited masters can transmit simultaneously
- Multiple masters can be present, but only one can control the bus at a time

- Only one master is allowed on the bus
- Two masters are allowed, but they must alternate

I2C supports a multi-master configuration, meaning multiple master devices can be connected to the same bus. However, only one master can control the bus at any given time. If two masters attempt to initiate communication simultaneously, the I2C protocol includes an arbitration mechanism to determine which master takes control, while the other backs off. This allows for flexible and coordinated communication in complex systems.

Correct answer

Question 11

1 / 1 pts

In UART communication, how are symbols synchronized?

- By synchronizing both devices to a global system clock.
- By using start and stop bits to explicitly frame each symbol and align the receiver's clock.
- By embedding a clock signal within the data stream.
- By sharing a clock line between the transmitter and receiver.
- By using edge detection on the data line to infer timing.

UART (Universal Asynchronous Receiver/Transmitter) communication uses explicit framing to achieve symbol synchronization. Each data frame begins with a start bit, which signals the receiver to begin sampling the incoming bits at predefined intervals based on the agreed-upon baud rate. The frame ends with one or more stop bits, marking the end of the symbol. This method allows the receiver to align its internal clock with the transmitter's data stream without needing a shared clock line, making synchronization explicit through framing.

Correct answer

Question 12

1 / 1 pts

What does the UART_SR register contain?

- Flags indicating the status of UART operations, such as transmit complete, receive buffer not empty, and error conditions.
- Control bits to enable or disable the UART peripheral.
- The actual data being transmitted or received.
- Configuration bits for setting baud rate, parity, and stop bits.
- Memory-mapped addresses for DMA transfers.

Answer Feedback: The UART_SR (Status Register) contains status flags that reflect the current state of the UART peripheral. These include flags like:

- TXE (Transmit Data Register Empty)
- RXNE (Receive Data Register Not Empty)
- TC (Transmission Complete)
- ORE (Overrun Error)
- FE (Framing Error)
- PE (Parity Error)

These flags are used by software to monitor and respond to UART events, such as when it's safe to write new data or when an error has occurred. The register does not contain configuration settings or data itself—those are handled by other registers like USART_CRx and USART_DR.

Correct answer

Question 13

1 / 1 pts

In serial communications, is symbol synchronization implicit or explicit?

- Implicit, because synchronization is handled entirely by hardware.
- Implicit, due to the use of fixed baud rates.
- Implicit, since the receiver guesses the timing based on signal transitions.
- Explicit, only asynchronous protocols use start and stop bits.
- Explicit, as it requires framing or control over the clock signal to align symbol timing.

Symbol synchronization in serial communications is explicit because it requires mechanisms to ensure that the receiver can correctly identify the boundaries of each symbol (bit or group of bits). In asynchronous protocols, this is achieved through framing—using start and stop bits to explicitly mark symbol boundaries. In synchronous protocols, the clock line explicitly controls when each symbol is valid. In both cases, synchronization is not inferred passively but is actively managed through protocol design, making it an explicit process.

Quiz Score: 7.5 out of 10

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